

Swarup Das

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Professional Summary

Fresh Computer Science graduate and immediate joiner with hands-on experience in full-stack and backend development, building production-ready REST APIs with authentication, caching, and containerization. Practical exposure to AI-integrated applications using LangChain and LLMs, with a strong foundation in Data Structures & Algorithms.

Education

Jalpaiguri Government Engineering College
Bachelor of Technology in Computer Science & Engineering

Nov 2022 – July 2026
CGPA: 7.7/10

Experience

Backend Developer Intern – Graphura India Private Limited  Jan 2026 – Apr 2026

- Developed and integrated **15+ REST APIs** using **Node.js** and **Express.js** for authentication, role-based access control, and frontend-backend communication.
- Collaborated with frontend developers using **GitHub** for feature integration and code reviews; designed optimized **MongoDB schemas** to improve query efficiency and backend scalability.

Projects

JOB TRACKER  | Node.js, TypeScript, Express.js, MongoDB, Redis, LangChain, Docker

- Built a full-featured **REST API** that helps job seekers track applications and leverage **AI (LangChain + Groq LLM)** to summarize job descriptions, score resume-to-job fit, detect scam postings, generate interview questions, and draft outreach messages.
- Designed a **chained AI pipeline** (prompt templates, output parsers, and sequential LangChain runnables) that processes resumes and job descriptions extracted from **PDF/DOCX uploads** via pdf-parse and mammoth, returning structured JSON insights.
- Implemented **Redis-based caching** for job lists, statistics, and AI responses to avoid redundant database and LLM calls, paired with **Redis-backed rate limiting** to enforce per-user API limits; automated 7-day follow-up reminders using **Nodemailer** and **node-cron**, and validated all 20+ endpoints using **Postman**, containerizing the full stack with **Docker** and **Docker Compose**.

STUDYBUDDY  | React.js, Node.js, Express.js, MongoDB, Socket.IO, Yjs, Redux, JWT

- Built a full-stack **real-time collaboration platform** that combines a **shared document editor** with team chat in one workspace, unlike standalone chat apps that require switching tools to share and edit content together.
- Engineered a **conflict-free collaborative text editor** using **Tiptap**, **Yjs**, and **y-websocket**, synchronized in real time across multiple concurrent users via **Socket.IO** event-based communication, alongside channel-based and 1-to-1 live chat with typing indicators and online presence tracking.
- Implemented **JWT authentication**, channel management (CRUD) with role-based member controls, and avatar uploads via **Cloudinary**; deployed the **React frontend on Vercel** and the **Node.js/Express backend on Render**.

INTERVO  | Next.js, TypeScript, Vapi AI, Groq LLM, Firebase, Zod, Tailwind CSS

- Built a voice-first **AI mock interview platform** that helps candidates practice **technical and behavioral interviews** through real-time voice conversations with role-specific, dynamically generated questions.
- Integrated **Vapi AI** for real-time voice interaction and **Groq LLM** for low-latency question generation and structured interview evaluation, enabling instant, conversational feedback after each session.
- Built with **Next.js (App Router, SSR)** for faster page loads and improved SEO over a traditional CSR setup, used **Zod** for end-to-end schema validation across forms and API routes, and stored user and interview data in **Firebase Firestore**.

Skills

Languages & AI: C++ , JavaScript, TypeScript, SQL, LangChain, Groq LLM, Vapi AI

Frontend: React.js, Next.js, Redux Toolkit, Tailwind CSS

Backend: Node.js, Express.js, REST APIs, MongoDB, Mongoose, Redis, JWT Auth, Socket.IO

DevOps & Tools: Docker, Git, Postman, Firebase, Vercel, Render, DSA, OOP, DBMS, OS, CN

Achievements

- Competitive Programming:** LeetCode rating **1700+** (top **12%**, **450+ problems solved**); **2-Star Coder** on CodeChef.
- Academic:** Secured **AIR 1905** in **WBJEE 2021** (top **1.3%**) out of 100,000+ candidates; cleared the **TCS NQT** and secured **All India Rank 2511** in **NCAT 2026** (Naukri Campus Aptitude Test).
- Research Internship:** Completed a research internship in **Quantum Computing** at **ISI Kolkata** (Summer 2023), studying quantum algorithms, quantum teleportation, and computational models under faculty guidance.